IDOM: Intent-Driven ODE for Multimodal Sequential Recommendation

Idea review · status after step 1 (no intent anchor yet) · 6 November 2026 · dataset: Amazon Beauty (FMRec protocol)

Summary

- **Idea.** Generate the next item in parallel, one flow trajectory per modality (ID, text, image), with a single shared vector field, and use a shared user intent to keep the modality trajectories consistent.
- **Step 1 result (ID + text, no anchor yet).** IDOM beats our reproduction of FMRec on all metrics over 3 seeds: **HR@10 9.18 vs 8.36 (+9.8%)**, NDCG@10 5.21 vs 4.97 (+4.8%), HR@20 +12.0%, NDCG@20 +6.7%.
- **Key finding.** In the FMRec design that IDOM inherits, the ODE trajectory is nominal. Inference with 1 Euler step gives the same accuracy as 30 steps (difference ≤ 0.01 points), and is 25× faster. The sampler only averages the per-step predictions, and the network almost ignores x_t and t.
- **Consequence.** As built, IDOM cannot claim to be faster than FMRec, and its "trajectory" story has no support yet. The gain comes entirely from fusing the ID and text predictions. The next steps are to make the trajectory matter, then add the intent anchor, and to compare against a baseline that fuses two separately trained models (ensemble).

1. Motivation and gap

- **Generative recommenders that use multiple modalities generate only in the ID space.** Other modalities enter only as conditioning signals (for example PrismRec, which uses one flow in the ID space with content as the condition). So preferences that live in a specific modality are never generated explicitly.
- **Flow-matching sequential recommenders** (FMRec, FlowRec) use only IDs. They claim efficiency from straight paths and few-step deterministic sampling.
- **Our measurement** shows that, in FMRec-style models, the number of steps does not change the ranking at all. The model behaves as a one-step predictor of the next item from the history. This is a new observation and can motivate the paper: *existing flow recommenders do not actually use their trajectory*.

Positioning against related work

Method	Generation space	Per-modality flow	Role of intent	What the flow is for
FMRec / FlowRec	ID	no	none	next item
PrismRec	ID (content as condition)	no	scalar gate (static vs dynamic)	next item
OT-FlowRec	fused	parallel flows, one per interest	prototypes define the flows	next item
UHIFlow	item features	yes (visual, text)	RQ codebook, does not guide the flow	uncertainty estimation
IDOM	**each modality**	**yes, shared vector field**	**shared anchor across modalities**	**next item**

The novelty claim must be narrower than "parallel multimodal flows are unexplored": UHIFlow and OT-FlowRec already have parallel flows, but for a different purpose (uncertainty) or along a different axis (interests).

2. Method (current implementation, built on FMRec)

- **One trajectory per modality.** The ID trajectory lives in the ID embedding space. The text trajectory lives in `text_proj(e_text)`, where e_{text} is a frozen sentence embedding (all-MiniLM-L6-v2) of the item's title, brand, categories and the first 64 words of its description.
- **One shared vector field.** All trajectories use FMRec's network. A learned modality embedding is added to the condition so the network knows which space it is in. The modalities are batched together (M·B rows), so there is one forward pass per step.
- **Training.** Each modality gets FMRec's objective in its own space: MSE on $\hat{x}_1$ + 0.2·CE + 0.4·auxiliary decoder loss. The losses are summed.
- **Inference.** Euler sampling for each modality, then fusion: score = Σ_m w_m·log_softmax($\hat{x}_m E_m^T$), with w = 1:1 for now.
- **Planned: intent anchor.** A training loss that aligns the per-modality $\hat{x}_1(t)$ on a shared intent within a range of t, plus inference-time guidance toward the intent at selected Euler steps (for example steps 1, 3, 5). This only makes sense once the trajectory actually affects the prediction (Section 3).

3. Why FMRec's ODE steps do not matter (code-level analysis)

Path and network. Training uses $x_t = t \cdot x_1 + (1-t) \cdot z_0$, $z_0 \sim N(0, I)$, and the network predicts $\hat{x}_1 = f_\theta(h, x_t, t)$. The state and the time enter the transformer only through:

```
rep_new = LayerNorm(history) + λ ⊙ (x_t + emb_t),     λ ~ N(0.001, 0.001²) per element
```

The cross-condition argument passed to the transformer blocks is unused. Inference uses Euler steps with $V = \hat{x}_1^{(i)} - z_0$.

1. **The sampler only averages the predictions.** Summing the N steps gives

   ```
   x_N = z0 + (1/N)·Σ_i (x̂1^(i) − z0) = (1/N)·Σ_i x̂1^(i)
   ```

   The noise z_0 cancels out, and more steps just means averaging more predictions. A standard rectified-flow update, $V = (\hat{x}_1 - x)/(1 - t)$, would correct errors from the current state. FMRec's update does not.
2. **The network barely sees x_t and t.** They are added at about 0.1% of the history scale, through multiplicative noise whose std equals its mean. So $\hat{x}_1^{(i)} \approx g(h)$ at every step.
3. **The objective has no use for x_t.** MSE + CE on $\hat{x}_1$ is minimised by a deterministic $E[x_1 \mid h]$, and top-k ranking does not need sampling. If x_t were injected strongly, the model could simply copy x_t when t is close to 1, which is probably why λ is kept tiny.

> **Implication.** FMRec is effectively a one-step sequence predictor, trained with tiny input noise, CE and an auxiliary decoder loss. Any IDOM claim about trajectories, step counts or anchors applied along the path needs a design in which x_t and t really influence the prediction.

4. Experiments so far

Amazon Beauty from FMRec's release (22,363 users, 12,101 items; identical to S3Rec's Beauty.txt). Leave-one-out split, full ranking, values in %, mean ± std over seeds {1997, 2024, 42}. Model selection follows FMRec: validation every 20 epochs, patience 5. The text features required recovering the item-to-ASIN mapping from the raw Amazon 2014 data; the mapping was verified on all 22,363 user sequences. "v3" = final code (noise drawn on GPU, top-k computed on GPU, checkpointing); the old and v3 code give the same accuracy.

4.1 FMRec reproduction and sanity check

Test	HR@10	NDCG@10	HR@20	NDCG@20	Seeds
FMRec (paper, Table 2)	8.27	4.95	11.63	5.79	–
FMRec, ours (old code)	8.41 ± 0.10	5.03 ± 0.04	11.88 ± 0.07	5.91 ± 0.03	2
FMRec, ours (v3)	8.36 ± 0.06	4.97 ± 0.02	11.71 ± 0.03	5.81 ± 0.03	3
IDOM, ID only (shared field + modality embedding)	8.30 ± 0.06	4.96 ± 0.04	11.81 ± 0.10	5.84 ± 0.05	3

The reproduction matches the paper. With a single modality, IDOM is equivalent to FMRec, so the shared-field architecture costs nothing.

4.2 Main result: IDOM with ID + text, no anchor (v3, test)

Test	HR@10	NDCG@10	HR@20	NDCG@20
FMRec (v3)	8.36 ± 0.06	4.97 ± 0.02	11.71 ± 0.03	5.81 ± 0.03
IDOM: ID trajectory only	8.41 ± 0.07	4.93 ± 0.07	11.82 ± 0.10	5.79 ± 0.08
IDOM: text trajectory only	7.02 ± 0.06	3.53 ± 0.04	10.93 ± 0.10	4.51 ± 0.05
IDOM: fused (ID + text)	**9.18 ± 0.10**	**5.21 ± 0.07**	**13.11 ± 0.12**	**6.20 ± 0.08**
Relative gain over FMRec	+9.8%	+4.8%	+12.0%	+6.7%

The gains are larger for HR and at @20: text mostly helps bring the target into the candidate set. An earlier run with the old code (seed 42) gave the same numbers (9.16 / 5.20).

- **The gain comes only from fusion.** The ID trajectory inside IDOM is no better than FMRec (8.41 vs 8.36).
- **Joint training weakens the text trajectory.** Validation HR@10 / NDCG@10 is 9.19 / 4.81 inside IDOM, versus 9.76 / 5.23 when text is trained alone. The shared field seems to favour the ID space.
- **So the obvious reviewer question is:** does IDOM beat simply fusing two separately trained models (FMRec + a text-only model)? This ensemble baseline is currently running.

4.3 Accuracy and time versus number of Euler steps (IDOM ID + text, test)

Steps N	Fused HR@10	Fused NDCG@10	ID HR@10 / NDCG@10	Text HR@10 / NDCG@10	Test pass time
1	9.17 ± 0.10	5.21 ± 0.06	8.41 / 4.93	7.03 / 3.53	4.4 s
2	9.18	5.21	8.41 / 4.93	7.02 / 3.53	8.0 s
5	9.17	5.21	8.41 / 4.93	7.02 / 3.53	19.3 s
10	9.17	5.21	8.41 / 4.93	7.02 / 3.53	37.0 s
30 (training default)	9.18	5.21	8.41 / 4.93	7.02 / 3.53	109.1 s

Per seed, the maximum difference across N is 0.01 points, far below the seed-to-seed spread (~0.1). FMRec checkpoints were evaluated in the same job; their numbers are not yet included in this note. The same behaviour is expected because the mechanism is identical.

4.4 Cost (single process, one GPU)

Model	Train s/epoch	Val pass, N=30	Val pass, N=1	Peak GPU memory	Total train time (v3, mean)	Stop epoch
FMRec	34.2	47.1 s	1.2 s	2.6 GB	1.9 h	300–440
IDOM (ID + text)	62.3	91.5 s	4.1 s	4.9 GB	4.8 h	420–500

The per-epoch and per-pass times come from a benchmark job on one node; the training-time column comes from the actual runs, possibly on different nodes. IDOM costs about 1.8–2× FMRec per epoch, linear in the number of modalities. With one Euler step, inference is about 20–40× cheaper for both models.

5. Assessment, risks and next steps

What holds

- The FMRec reproduction is faithful, and the shared-field architecture is lossless with a single modality.
- Generating in the text space adds a consistent gain: about 10× the seed std on HR@10.
- The observation "the flow trajectory is nominal" is reproducible and can motivate the paper.

What does not hold yet

- **Speed claim:** IDOM is not faster than FMRec, and FMRec can also run with 1 step.
- **Trajectory and anchor story:** not supported while x_t and t are ignored by the network.
- **Contribution beyond ensembling:** unproven until IDOM beats the separate-model ensemble.

Planned next steps

1. **Ensemble baseline** (running): text-only models with checkpoints, then fusion with FMRec, fusion weight tuned on validation.
2. **Diagnostics:** train with $\lambda \in \{0, 0.1, 1.0\}$, and measure how much $\hat{x}_1$ changes with x_t and t. If $\lambda = 0$ matches FMRec, the flow component contributes nothing.
3. **Make the trajectory matter:** inject x_t as its own token or via AdaLN on t, and use the state-dependent sampler $V = (\hat{x}_1 - x)/(1 - t)$. The target is accuracy that improves with steps and stays high with few steps.
4. **Intent anchor:** a training-time alignment of the per-modality $\hat{x}_1(t)$, plus inference-time guidance at selected steps. Success criteria: beat the ensemble baseline, recover the text trajectory (val HR@10 9.19 → 9.76 or higher), and keep accuracy with few steps.
5. **Scale-up:** datasets with images and text (for example Amazon Sports/Toys, MicroLens), and baselines (PrismRec, OT-FlowRec, DMMD4SR, DiffuRec, DreamRec) measured with the same harness, including time per epoch and per inference.

> #### Questions for the reviewer
>
> 1. **Framing:** (a) "flow recommenders ignore their trajectory; IDOM makes it matter through intent-guided multimodal trajectories", or (b) "a one-step multimodal generator trained by flow matching, with intent alignment". (a) is stronger but riskier.
> 2. **Evidence bar:** is the separate-model ensemble the right baseline to beat, alongside PrismRec and OT-FlowRec?
> 3. **Scope:** given the deadline, should we restrict to ID + text first and add images only if time allows?